RobustOps: A Model-Agnostic Framework for Distribution-Shift and Adversarial Risk Assessment of AI Systems

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Abstract

Production ML systems fail under both natural distribution shift and adversarial perturbation, yet these risks are usually handled by separate tools and mitigation strategies. We present **RobustOps**, a unified, model-agnostic framework structured around four pillars—Diagnose, Score, Mitigate, and Observe—and aligned with the NIST AI Risk Management Framework. Its central idea is shift-type-aware mitigation selection using the $P(X, Y)$ decomposition, motivated by the observation that incorrect mitigation can be worse than no action. Empirically, replay-based continual learning under concept drift causes a 0.25 R^2 drop, while adversarial training under pure covariate shift reduces accuracy without improving robustness. We also replace a linear risk score with a diagnosis-aware two-stage gate that supports four operational decisions: PASS, MONITOR, ABSTAIN, and BLOCK. For grounded documentation, we add a retrieval-augmented reporting layer over NIST AI RMF and AMLAS with citation verification. Across image, NLP, and tabular settings, the framework supports cross-modality diagnosis, mitigation routing, and interpretable audit outputs, and is demonstrated through an interactive Streamlit dashboard.

1. Introduction

Machine learning systems deployed in production fail for two structurally different reasons. The first is *natural distribution shift*: the data encountered at inference time diverges from the data used during development, whether through covariate drift, concept change, or label-prior shift (Moreno-Torres et al., 2012; Huyen, 2022; Paleyes et al., 2022). Surveys of deployed ML systems consistently identify distribution shift as a major cause of post-deployment degradation (Paleyes et al., 2022; Sculley et al., 2015). The second is *adversarial perturbation*: an attacker crafts inputs designed to induce incorrect predictions within a bounded perturbation budget (Madry et al., 2018; Goodfellow et al., 2015). These failure modes are typically studied and managed separately.

Existing tooling reflects this split. Monitoring platforms such as Evidently, Alibi Detect, NannyML, and Arize surface drift metrics; robustness benchmarks such as RobustBench (Croce et al., 2021) report attack success rates; continual learning surveys (van de Ven et al., 2022; Wang et al., 2024) organize adaptation strategies by taxonomy. To our knowledge, no existing tool or published framework selects mitigation based on a *diagnosed* failure mode that distinguishes covariate, label, and concept shift from adversarial perturbation within one pipeline. RobustOps is intended to fill this integration gap.

This distinction matters because the wrong mitigation can be worse than none. Replay-based continual learning (Buzzega et al., 2020; Lesort et al., 2021) approximates joint training over old and new data and implicitly assumes the underlying mapping $P(Y|X)$ is stable. Under concept drift, this assumption fails and replay can anchor the model to stale relationships. We show in Section 5 that this produces a 25-point R^2 degradation on California Housing. Symmetrically, adversarial training (Madry et al., 2018) trades clean accuracy for robustness (Tsipras et al., 2019); when applied to pure covariate shift with no adversarial threat, it pays this cost without benefit. Without diagnosis, a practitioner cannot reliably distinguish these cases from surface-level drift metrics alone.

We introduce **RobustOps**, a model-agnostic framework organized around four pillars—**Diagnose**, **Score**, **Mitigate**, and **Observe**—that map naturally to identify, detect, respond, and recover stages of operational AI defense, and align with the Measure and Manage functions of the NIST AI Risk Management Framework (NIST, 2023). The framework emphasizes resilience, graceful degradation, explainability, and verifiability through a diagnosis-first decision

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process.

Hypothesis. We test the following claims:

- **H1 (Primary).** Shift-type-aware mitigation selection, guided by the decomposition of $P(X, Y)$ into covariate, label, concept, or mixed shift, improves post-mitigation performance relative to fixed or generic mitigation.
- **H2 (Negative).** Applying mitigation strategies misaligned with the underlying shift type—for example, replay-based continual learning under concept drift or adversarial training under pure covariate shift—causes degradation relative to no intervention.
- **H3 (Attribution).** Modeling drift attribution as a multiple-instance learning problem improves identification of drift-contributing samples compared to per-feature statistical baselines. We test H3 in Section 5; the strong form is not supported.
- **H4 (Gating).** A diagnosis-aware gating policy that combines bounded nonlinear severity with a shift-type-indexed action table produces more operationally meaningful decisions than a linear weighted-score threshold.

H2 is the strongest empirical anchor: Experiments 2a and 2b directly test whether incorrect mitigation is harmful. H1 is supported through contrast with the abstain branch, H3 through comparative attribution experiments, and H4 through the diagnosis-aware gate sweep.

Contributions. [1] We make five contributions:

1. A four-pillar framework unifying distribution-shift and adversarial risk assessment under a single diagnosis-first pipeline, aligned with NIST AI RMF and AMLAS (Hawkins et al., 2021).
2. Shift-type-aware mitigation selection via the $P(X, Y)$ decomposition, with empirical evidence that misapplied mitigation is worse than no action.
3. A cross-modality evaluation on image, NLP, classification, and regression settings, showing diagnoses spanning four shift categories.
4. A grounded compliance-reporting layer using hybrid retrieval, reranking, and citation verification to prevent hallucinated references.
5. A diagnosis-aware gating policy and a comparative attribution study showing that distance-based methods outperform learned attention-MIL variants under additive-noise drift.

[1] Code, dashboard, and reproducibility artifacts: https://github.com/CynthiaK1/INSE-6450/tree/main

2. Background and Related Work

Distribution shift taxonomy. The decomposition $P(X, Y) = P(Y|X)P(X) = P(X|Y)P(Y)$ separates shift into three canonical types: covariate shift ($P(X)$ changes, $P(Y|X)$ stable), label shift ($P(Y)$ changes, $P(X|Y)$ stable), and concept drift ($P(Y|X)$ changes) (Moreno-Torres et al., 2012). Huyen (2022) emphasizes that the appropriate response depends on the shift type. Rabanser et al. (2019) show empirically that no single detector dominates across all regimes.

Continual learning under shift. The stability–plasticity dilemma frames continual learning as the balance between adapting to new data and preserving prior knowledge (van de Ven et al., 2022). Methods are commonly grouped into regularization-based, replay-based, and architectural families (Kirkpatrick et al., 2017; Buzzega et al., 2020). Replay is widely applicable, but prior work does not generally condition its use on diagnosed shift type.

Adversarial robustness. Bounded ℓ_∞ adversarial examples remain the standard robustness threat model (Madry et al., 2018). AutoAttack (Croce and Hein, 2020) is a strong evaluation standard but can be expensive for CI/CD contexts; FGSM (Goodfellow et al., 2015) is lighter-weight. Tramèr et al. (2020) show that many defenses fail under adaptive attack, motivating conservative abstention when the threat exceeds declared assumptions. Tsipras et al. (2019) show that adversarial training trades clean accuracy for robust accuracy.

AI risk frameworks and MLOps. The NIST AI Risk Management Framework (NIST, 2023) organizes AI governance into Govern, Map, Measure, and Manage. ISO/IEC 42001 (ISO, 2023) complements this with management-system guidance. AMLAS (Hawkins et al., 2021) contributes a structured assurance perspective. Broader deployment challenges are discussed by Paleyes et al. (2022), Sculley et al. (2015), and Breck et al. (2017).

Multiple-instance learning and attribution. Attention-based MIL (Ilse et al., 2018) provides instance-level relevance within bag-level predictions. This structure motivates its use for drift attribution, where a flagged evaluation window can be treated as a bag and candidate drifted samples as instances.

Positioning. Existing work addresses shift detection, continual learning, adversarial robustness, and monitoring largely as separate subfields. RobustOps combines these under a diagnosis-first pipeline that explicitly links diagnosis to mitigation selection.

3. The RobustOps Framework

3.1. Threat Model

We state the threat model explicitly because later design choices depend on it.

Model access. Grey-box by design. The framework consumes the model artifact and a *reference buffer* of historical labeled samples, but not necessarily the original training set. In our experiments the reference buffer comes from public benchmark training splits, but the interface is intentionally restricted so the same pipeline can audit third-party or regulated models.

Natural shift. Unknown-type distribution shift in $P(X, Y)$, decomposed into covariate, label, concept, or mixed shift.

Adversarial threat. Bounded ℓ_∞ perturbation with budget ε, applied at inference time by a white-box attacker with gradient access. We do **not** claim robustness to adaptive attackers (Tramèr et al., 2020) or attacks beyond the declared budget.

Evaluation context. Offline, CI/CD-style pre-deployment gate rather than real-time defense.

Out of scope. Data poisoning, backdoors, supply-chain attacks on the model artifact, and privacy attacks.

3.2. Pipeline Overview

The framework's four pillars instantiate the core operational defense cycle: **Diagnose** identifies and detects risk; **Score** determines severity and gate action; **Mitigate** selects an appropriate response; and **Observe** supports recovery through attribution, auditability, and human review. Figure 1 shows the full pipeline.

Why model-agnostic. The framework depends only on a forward pass, optional gradients for FGSM, and a reference buffer. It does not require architecture internals or access to the original training process. The Mitigate pillar is where modality-specific implementation details appear.

3.3. Pillar 1: Diagnose

The Diagnose pillar runs three parallel analyses on each evaluation window W.

Drift detection. Population Stability Index (PSI) is computed per feature; drift is flagged when PSI exceeds 0.2 on at least two features for at least two consecutive windows.

Shift-type classification. Given a historical buffer and a new window, the framework computes:

- $D_{P(X)}$: Maximum Mean Discrepancy with RBF kernel between buffer and window features.
- $D_{P(Y|X)}$: Jensen–Shannon divergence between prediction distributions for classification, or normalized prediction-mean shift for regression.
- $D_{P(Y)}$: Jensen–Shannon divergence between label marginals.

The shift type is then classified as *concept* if $D_{P(Y|X)}$ dominates, *label* if $D_{P(Y)}$ dominates, *covariate* if $D_{P(X)}$ dominates, *mixed* otherwise, and *none* if all values remain below threshold.

Adversarial stress test. A modality-appropriate FGSM attack generates perturbed inputs, and the Diagnose pillar reports adversarial success metrics such as accuracy drop and worst-case degradation.

3.4. Pillar 2: Score

The Score pillar combines Diagnose outputs into a bounded severity indicator and a diagnosis-aware gate decision. Severity is computed with a noisy-OR aggregation:

$$R^* = 1 - \prod_{i=1}^{4}(1 - r_i),$$

where $r_i \in \{\Delta_{\text{acc}}, \Delta_{\text{conf}}, \text{ASR}, \text{WCASR}\}$ are clipped to $[0, 1]$. Compared with a linear weighted sum, this formulation is bounded, captures interactions between risk components, and saturates more naturally when multiple risks co-occur. We map R^* to four severity tiers (`none`, `low`, `moderate`, `high`) and emit one of four actions: PASS, MONITOR, ABSTAIN, or BLOCK.

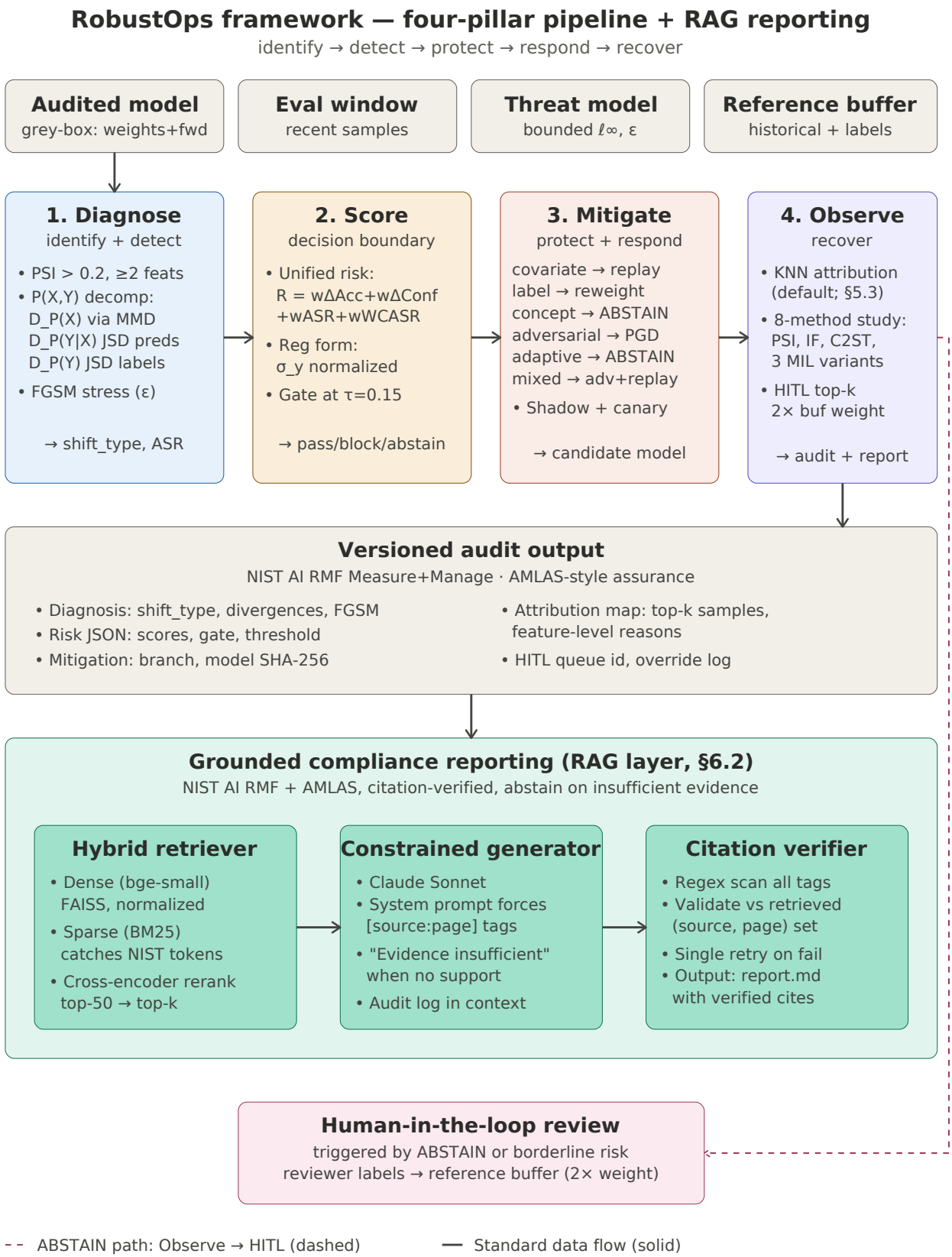
For regression, we use a separate severity formulation:

$$R^*_{\text{reg}} = w_1(R_c^2 - R_p^2) + w_2 \frac{|\text{RMSE}_p - \text{RMSE}_c|}{\max(\sigma_y, \delta)} + w_3 \cdot \text{ASR}_{\text{reg}},$$

where σ_y is the target standard deviation and δ is a small floor.

3.5. Pillar 3: Mitigate

The Mitigate pillar is the framework's core operational contribution: a decision table that links diagnosed conditions



Threat model: §3.1 · pillar instantiations: §3.3–3.6 · NIST/AMLAS alignment: §3.7 · RAG layer: §6.2

Table 1: Mitigation decision table indexed by diagnosed condition. Cases marked **abstain** route to human review rather than automatic action.

Diagnosis	Mitigation
Covariate, mild	Replay-CL, 80/20 mix
Covariate, severe	Replay-CL + adversarial fine-tuning
Label	Prior correction (no retrain)
Concept	**Abstain and alert**
Adversarial, bounded	Adversarial training (PGD on buffer)
Adversarial, adaptive	**Abstain**
Mixed	Adversarial training on replay buffer
Unknown / low confidence	**Abstain** (route to Observe)

to appropriate responses. Table 1 summarizes the routing policy.

The key idea is that mitigation should depend on shift type. Replay-based continual learning assumes a stable $P(Y|X)$ and can fail under concept drift, while adversarial training is specifically designed for adversarial threats and can be wasteful or harmful outside that regime.

3.6. Pillar 4: Observe

The Observe pillar provides sample-level attribution and human-in-the-loop review. Given a flagged window, the framework identifies which samples contributed most to the drift signal. We evaluate multiple attribution families: feature-based statistics, distance-based methods, anomaly detectors, discriminative two-sample classifiers, and three learned MIL variants.

The attention-MIL framing is structurally appealing because a flagged window behaves like a bag with a bag-level drift label, and per-sample attention offers an instance-level relevance score (Ilse et al., 2018). However, our empirical results show that under additive isotropic Gaussian drift, distance-based methods significantly outperform the learned MIL variants. We therefore adopt **KNN distance to the reference buffer as the default attribution signal**.

Grounded compliance reporting layer. The Observe pillar also exposes a retrieval-augmented generation layer that converts audit logs into NIST AI RMF and AMLAS-aligned compliance text without hallucinated citations. Standards documents are chunked, embedded, indexed with both dense retrieval and BM25, reranked with a cross-encoder, and then passed to a constrained generator. Every factual claim must cite retrieved evidence, and unsupported citations trigger verification failure and retry.

HITL integration. Top-ranked samples are surfaced to human reviewers along with feature-level explanations. Reviewer feedback is then weighted into the replay buffer, allowing limited annotation effort to have targeted downstream impact.

4. Methodology

Models and datasets. We evaluate ResNet-18 on CIFAR-10, DistilBERT on SST-2, XGBoost classification on Adult Income, and XGBoost regression on California Housing.

Drift protocol. We generate three sequential drift events per modality with increasing severity. For tabular data we use Gaussian noise, for images normalized-pixel noise, and for NLP embedding-space perturbations. The third event additionally introduces a 15% label-flip rate.

Adversarial protocol. We use FGSM with $\varepsilon = 0.3$ for tabular, $8/255$ for image, and character-level perturbation for NLP.

Mitigation experiments. We focus on two controlled validation cases: (2a) replay-based continual learning on concept drift, expected to fail, and (2b) adversarial training under pure covariate shift, expected to waste compute and reduce accuracy.

Attribution methods. We compare eight attribution methods plus a hybrid. Evaluation windows contain 500 samples, including 50 drifted points, and performance is measured with recall@50 over 50 trials using paired Wilcoxon signed-rank tests and Cohen's d.

Diagnosis-aware gate sweep. We evaluate the gate across all 24 combinations of severity tier and diagnosed shift type, with an ambiguity override threshold of $\delta = 1.5$.

5. Results

5.1. Cross-Modality Diagnosis

Table 2 reports the $P(X, Y)$ decomposition across four modalities and three drift events. The results exercise four distinct diagnosis branches: tabular classification spans *none* $\rightarrow$ *concept* $\rightarrow$ *label*; tabular regression is consistently *concept*; image transitions from *none* to *concept*; and NLP remains *covariate*.

5.2. Mitigation Toolbox Validation

Experiment 2a: Replay on concept drift fails. We applied replay-based continual learning with an 80/20 buffer mix to the California Housing concept-drift case. The baseline regressor achieved $R^2 = 0.69$ on original test data; after replay retraining it achieved $R^2 = 0.44$ on the drifted distribution, a 25-point degradation. This supports H2: under concept drift, the replay buffer preserves stale feature–target relationships that actively harm performance.

Table 2: Cross-modality $P(X, Y)$ decomposition diagnoses. Bold cells indicate the dominant divergence component.

Mod.	Drift	$P(X)$	$P(Y\|X)$	$P(Y)$	Dx
tab_cls	1	0.005	0.002	0.000	none
tab_cls	2	0.005	**0.016**	0.000	conc
tab_cls	3	0.005	0.008	**0.011**	label
tab_reg	1	0.091	**0.992**	0.049	conc
tab_reg	2	0.106	**1.089**	0.080	conc
tab_reg	3	0.107	**1.290**	0.033	conc
image	1	0.006	0.002	0.002	none
image	2	0.010	**0.017**	0.004	conc
image	3	0.026	**0.066**	0.003	conc
nlp	1	**1.616**	0.000	0.000	cov
nlp	2	**1.903**	0.000	0.000	cov
nlp	3	**1.911**	0.000	0.001	cov

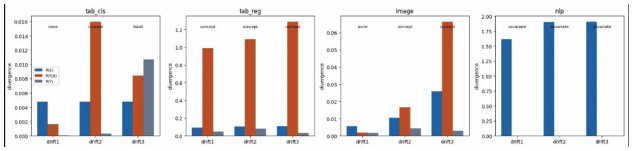


Figure 2: Cross-modality $P(X, Y)$ decomposition diagnoses across four modalities and three drift events. Tabular classification transitions from none to concept to label; tabular regression is concept-dominated; image shifts from none to concept; NLP remains covariate.

Experiment 2b: Adversarial training wastes compute on pure covariate shift. We trained baseline and adversarially trained MLPs on Adult Income. Baseline clean accuracy was 0.856; adversarial training reduced it to 0.852. Under pure covariate shift, baseline accuracy was 0.738 while the adversarially trained model reached 0.710. This again supports H2: a mitigation designed for the wrong threat can degrade both clean and shifted performance.

Together, these experiments validate the framework's central claim that mitigation selection should be conditioned on diagnosis.

5.3. Sample-Level Attribution: Comparative Study

H3 predicted that multiple-instance learning would improve identification of drifted samples over simpler baselines. We tested this against several baselines using recall@50 over 50 trials. Table 3 reports the results.

KNN distance achieved the strongest standalone performance. All three learned MIL variants underperformed distance-based methods significantly. H3 in its strong form is therefore not supported. Under additive isotropic Gaussian drift, this is intuitive: Euclidean distance is well matched to the geometry of the perturbation. Still, the 50/50 KNN+InstanceMIL hybrid statistically tied pure KNN, suggesting the learned signal may be complementary rather than redundant.

Table 3: Sample-level drift attribution performance measured by recall@50 over 50 trials under additive Gaussian drift on Adult.

Method	Mean	Std
Hybrid (KNN+InstanceMIL, λ=0.5)	**0.904**	0.034
KNN distance (default)	0.903	0.028
Mahalanobis distance	0.742	0.059
Isolation Forest	0.694	0.054
InstanceMIL (max-pool, BCE)	0.538	0.073
Attention-MIL (softmax, BCE)	0.469	0.073
Classifier two-sample test	0.379	0.071
Per-feature PSI	0.259	0.109
Sparsemax MIL	0.211	0.045

5.4. Diagnosis-Aware Gating Policy

The original RobustOps gate used a linear weighted risk score and emitted PASS, BLOCK, or ABSTAIN based on a threshold. In practice, this formulation proved too coarse: across ablations it was threshold-dominated and failed to distinguish operationally different situations.

We replace it with a two-stage diagnosis-aware policy function $G = g(R^*, \hat{s}, D_{P(X)}, D_{P(Y|X)}, D_{P(Y)})$, where $\hat{s}$ is the diagnosed shift type. The action space is extended to {PASS, MONITOR, ABSTAIN, BLOCK}.

Stage 1: noisy-OR severity. The four risk components are aggregated into

$$R^* = 1 - \prod_{i=1}^{4}(1 - r_i),$$

and mapped to four severity tiers: `none`, `low`, `moderate`, and `high`.

Stage 2: policy lookup. The gate then indexes a hand-specified policy table by severity tier and diagnosed shift type. Concept drift routes to ABSTAIN at every non-zero severity, adversarial cases escalate faster than covariate ones, and label shift is treated more permissively.

Confidence override. If no divergence component clearly dominates, the gate returns ABSTAIN regardless of severity. This fail-safe behavior prevents confident action under ambiguous diagnosis.

Figure 4 shows that the revised gate exercises all four actions across the input space, unlike the original linear formulation. Its decisions are also locally inspectable, which is advantageous under assurance-oriented frameworks such as AMLAS.

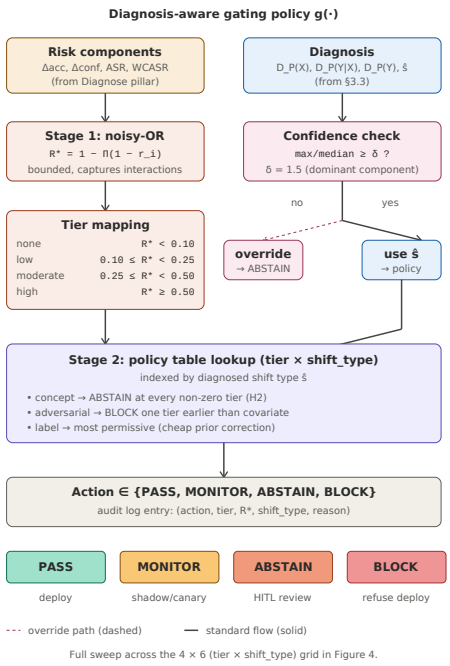


Figure 3: Diagnosis-aware gating policy. Risk components are aggregated into a bounded severity R^* via noisy-OR and mapped to four tiers; diagnosis confidence is checked in parallel. Ambiguous diagnoses trigger ABSTAIN, while confident (tier, shift-type) pairs map to PASS, MONITOR, ABSTAIN, or BLOCK.

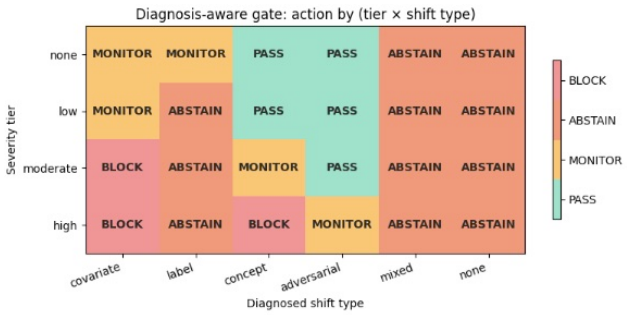


Figure 4: Diagnosis-aware gate outputs across the 4×6 grid of severity tiers and diagnosed shift types. Concept drift routes to ABSTAIN at every non-zero tier, adversarial cases escalate earlier than covariate cases, and ambiguous diagnoses trigger the fail-safe override.

6. Grounded Compliance Reporting

An interactive Streamlit dashboard reads the framework's JSON/CSV outputs and surfaces the Diagnose, Score, Mitigate, and Observe pillars, gate history, and NIST AI RMF mappings. We treat this as a demonstration layer rather than a standalone research contribution.

To convert audit logs into NIST AI RMF and AMLAS-aligned compliance documents without hallucinated citations, we implement a four-stage retrieval-augmented generation pipeline. First, standards documents are chunked with sentence-boundary-aware splits and embedded into a dense index, with a parallel BM25 sparse index. Second, dense and sparse retrieval run jointly, and top candidates are reranked with a cross-encoder. Third, a constrained generator produces report text in which every factual claim must end with a citation tag drawn from retrieved evidence. Fourth, a verifier checks all emitted citations against the retrieved set and forces retry if unsupported citations appear.

This design addresses a practical problem in LLM-generated compliance writing: fabricated citations are worse than no citations because they create false confidence. In our demonstration, the verifier accepted all generated sections without fabricated references and correctly produced "Evidence insufficient" when retrieval did not support a requested claim.

7. Limitations and Future Work

Evaluation scope. We validate on ResNet-18, DistilBERT, and XGBoost. Extending evaluation to larger vision transformers, LLM-based systems, and broader deployment contexts is future work. The HITL module uses ground-truth labels as a proxy for human review, so annotation noise and latency are not modeled.

Framework coverage. The mitigation table is validated on two representative controlled cases rather than exhaustively across all regimes. The attribution study is specific to additive isotropic Gaussian drift, where Euclidean distance is well matched to the perturbation geometry. Extending the Observe pillar to more structured drift types is a priority.

Threat scope. We do not claim robustness against adaptive attackers (Tramèr et al., 2020) or attacks beyond the declared ε budget. Similarly, shadow/canary deployment integration is described architecturally but not fully implemented.

Artifact integrity. Our threat model focuses on inference-time perturbation, but the model artifact itself is also an attack surface. PyTorch's `torch.load`, for example, relies on Python pickle and has known security concerns. A complete risk-assessment system should therefore verify artifact provenance and integrity before any of the four pillars run.

8. Conclusion

We presented RobustOps, a unified four-pillar framework for distribution-shift and adversarial risk assessment of AI systems. Its central contribution is shift-type-aware mitigation selection based on the $P(X, Y)$ decomposition, supported by experiments showing that incorrect mitigation can be measurably harmful. We further introduced a diagnosis-aware gating policy, a grounded compliance-reporting layer with citation verification, and a comparative attribution study in which distance-based methods outperform learned MIL variants under additive-noise drift. Overall, RobustOps connects diagnosis, mitigation, attribution, and auditability within a single model-agnostic pipeline aligned with NIST AI RMF and AMLAS.

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